Evaluating Comparable-Sales and Machine Learning Approaches for House Price Prediction

By: Jenny Nguyen Ximena Chen, Kai Binatti

CSC542: Statistical Learning with Applications
Professor Vanessa Aguiar-Pulido

Final Submission | April 2026

Agenda

- Introduction
- State-of-the-art
- Materials and Methods
- Evaluation
- Results
- Conclusion and Future Work

Introduction

Prices rising faster than income

Record low share of new buyers

Monthly payments increasingly unaffordable

Household income has stagnated

KNN vs Random Forest -
which model prices home better?

$360,591
Median Home Price
Feb 2026 (Zillow)

21%
First-Time Buyers
Historic low (NAR, 2025)

6.37%
30-yr Mortgage Rate
April 2026 (Freddie Mac)

$83,730
Median Household Income
2024, largely flat (Census)

Introduction

Traditional: Comparable Sales (Comps)

- Find recently sold homes that look similar in similar in size, location, and features. Average their prices to estimate the target home's value

Our Approach: KNN vs Random Forest

- KNN regression models the comps logic computationally. Random Forest uses 500 decision trees to capture complex patterns. We compare both across tiers

Research Question: Can Random Forest outperform KNN for home valuation and where does each model struggle?

State-of-the-art

Industry already uses ML at scale

Ensemble models beat traditional ones

KNN mimics comps but is fragile

Luxury homes remain hardest to value

Gap we Investigate
Where do models fail? Price tier analysis

Automated Valuation Models

Zillow's Zestimate runs 7.5M+ machine learning models per property.
Captures patterns comparable sales cannot.

Ensemble Models Dominate

Random Forest and Gradient Boosting beat Support Vector Machine on Mean Squared Error, Root Mean Squared Error and Mean Absolute Percentage Error across 40,000+ transactions (Ho et al., 2021).

K-Nearest Neighbors as a Comps Proxy

Finds similar homes, averages their price.
Breaks down when comparable sales are sparse.

The Luxury Gap

Both models struggle at the high end.
Unique homes lack comparable sales — a known gap in the literature.

Dataset and Feature Selection

The Dataset

79 variables

2930 observations

Cleaning The Data

>4000 sq.ft living area - Outliers

Categorical Missing Values - “None”

Feature Selection

Hedonic Pricing Theory - value is the sum of attributes. 4 Critical “Pillars”:

Size: Total Living Area (**TotalSF**) and Land Size (**LotArea**)

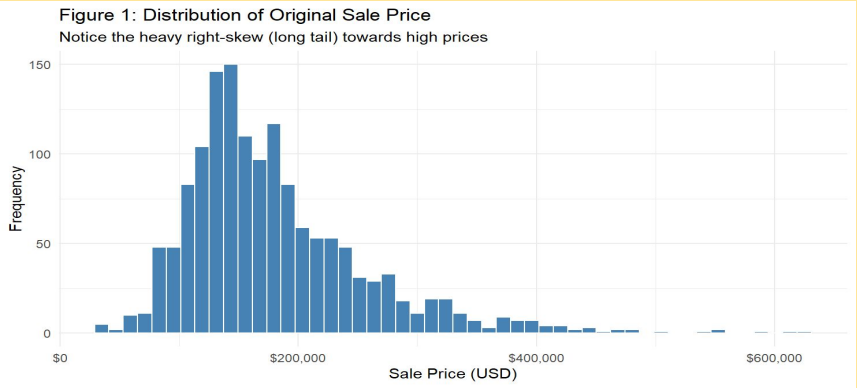
Quality: Overall Structural Grade (**OverallQual)** and Kitchen Renovation Level(**KitchenQualNum)**

Age: (HouseAge): year built - year sold

Utility & Location: Bathroom Count (**TotalBath),** Garage Capacity (**GarageCars)** and a neighborhood- based price proxy (NB_MeanLogPrice).

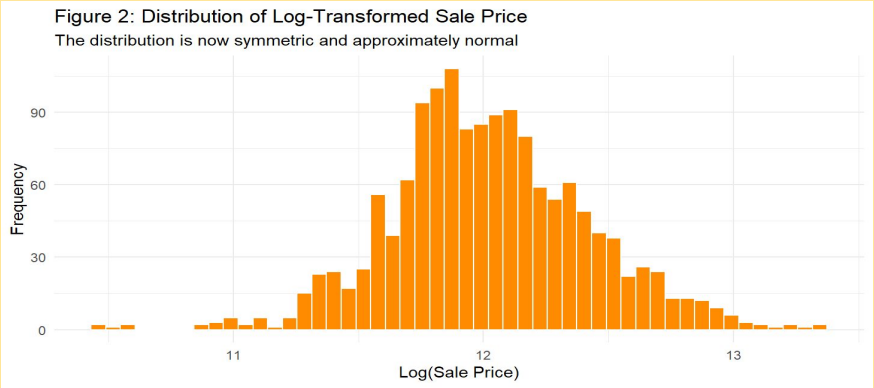
Target Variable Transformation

Original Sale Price - right skewed
Could lead to **heteroscedasticity-** errors grow disproportionately with price

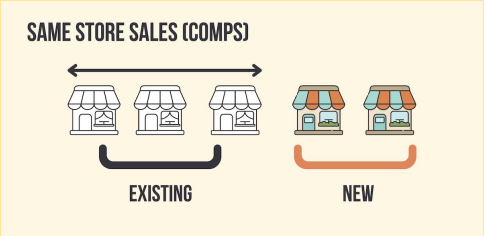


Most homes - $100k-200k, with a long tail until $600k

The Solution?
Log transformation to stabilize variance



The KNN regression Model

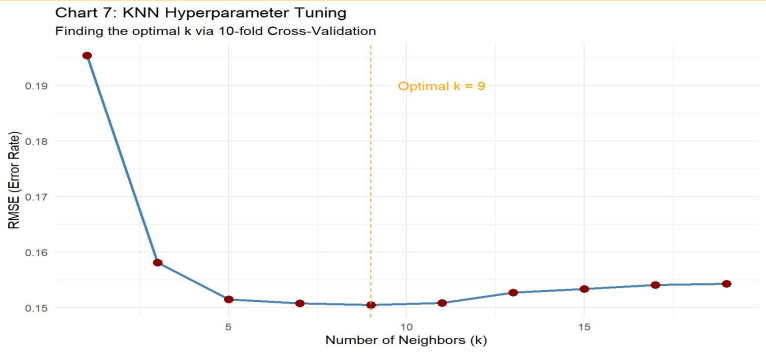


8 core features: TotalSF, OverallQual, TotalBath, HouseAge, LotArea, GarageCars, KitchenQualNum, NB_MeanLogPrice.

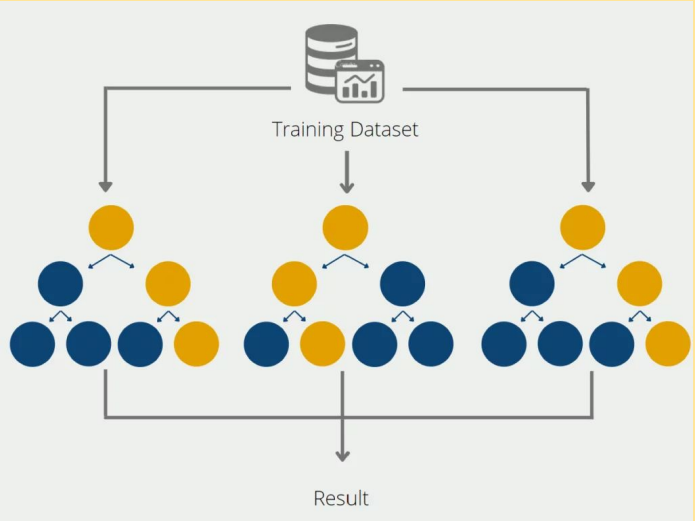
K-Nearest Neighbors to mimic the “Comparable Sales” logic, using **Euclidean Distance**

We used **10-fold Cross Validation** to determine the **optimal K value**

All features were standardized using **z-score normalization** to avoid domination of certain observations.



The Random Forest Model



Number of trees = 500

Model can predict, for example, that a high-end kitchen adds significantly more value to a new house than to a distressed older property.

Trained using **mtry (Square root of the number of predictors),** allowing trees to grow to their maximum depth.

Evaluation

Evaluation Framework:

- **Hold-out Method:** Dataset partitioned into 80% training set; 20% validation sets

- **Multi-Metric Assessment:** Result was evaluated within three lenses:
  - **Log-RMSE -** For mathematical precision and error penalty
  - **MAE -** To understand the average dollar-value deviation in a real-world context
  - **MAPE -** To provide intuitive accuracy measures

- **Tiered Stratification:** Models were evaluated across three price tiers: **Low-End, Mid-Range, Luxury**

to identify where algorithmic valuation succeeds and where it requires human intervention.

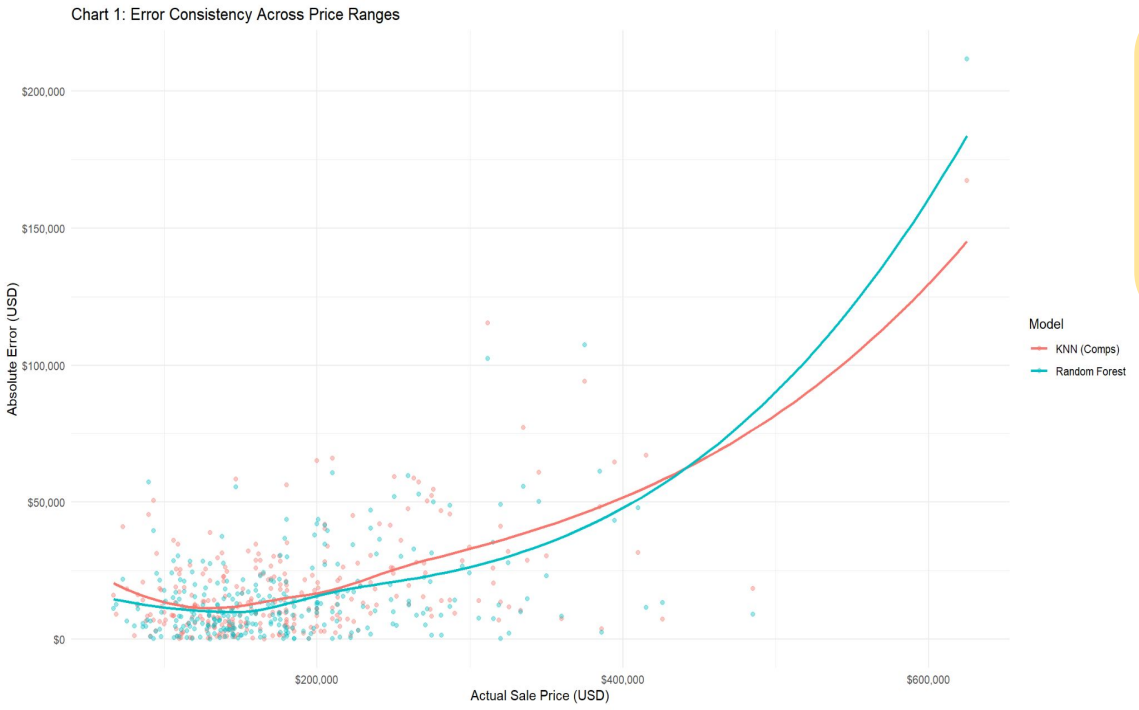
Results

Overall Model Performance

- **Strong Linear Relationship (Points close to the diagonal)**
- **Random Forest: Higher Accuracy**

Results

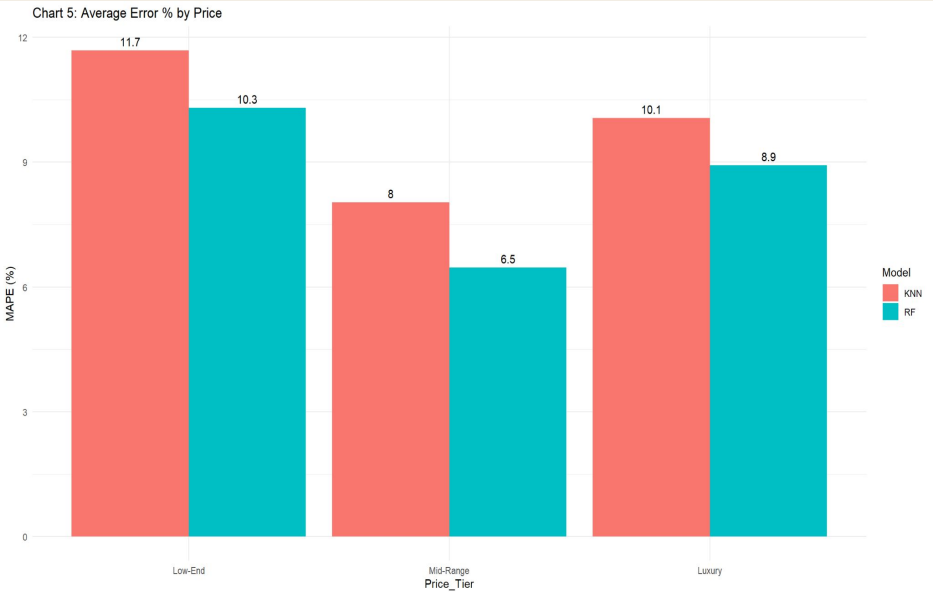
Error Analysis: how prediction changes as house price increases

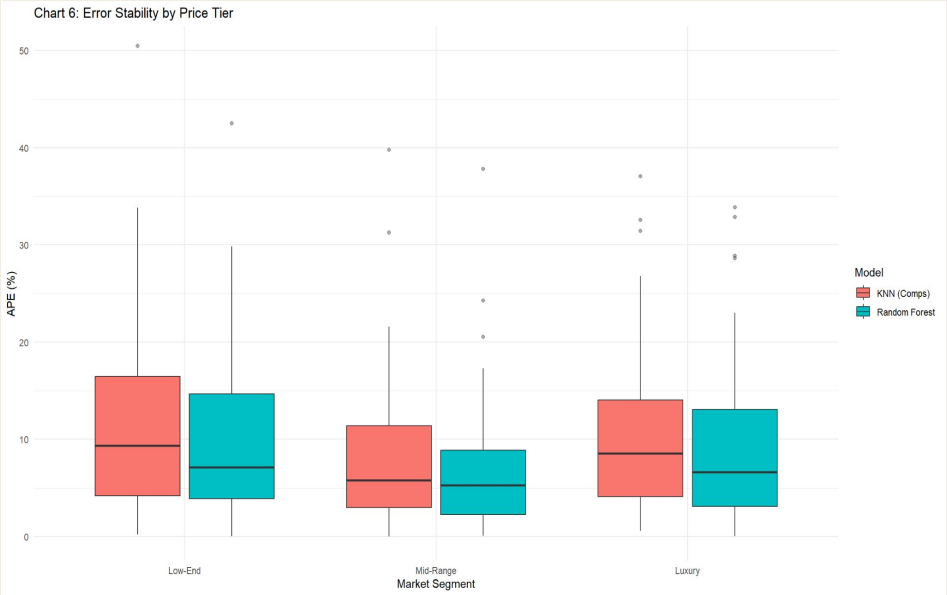


- **Lower & Mid Price Level (small and stable)**
- **Above $450,000 (error increases)**
- **Expensive House Unique**
- **Random Forest Lower Error (more reliable)**

Results

Segmented Performance





Best Performance in Mid-Range

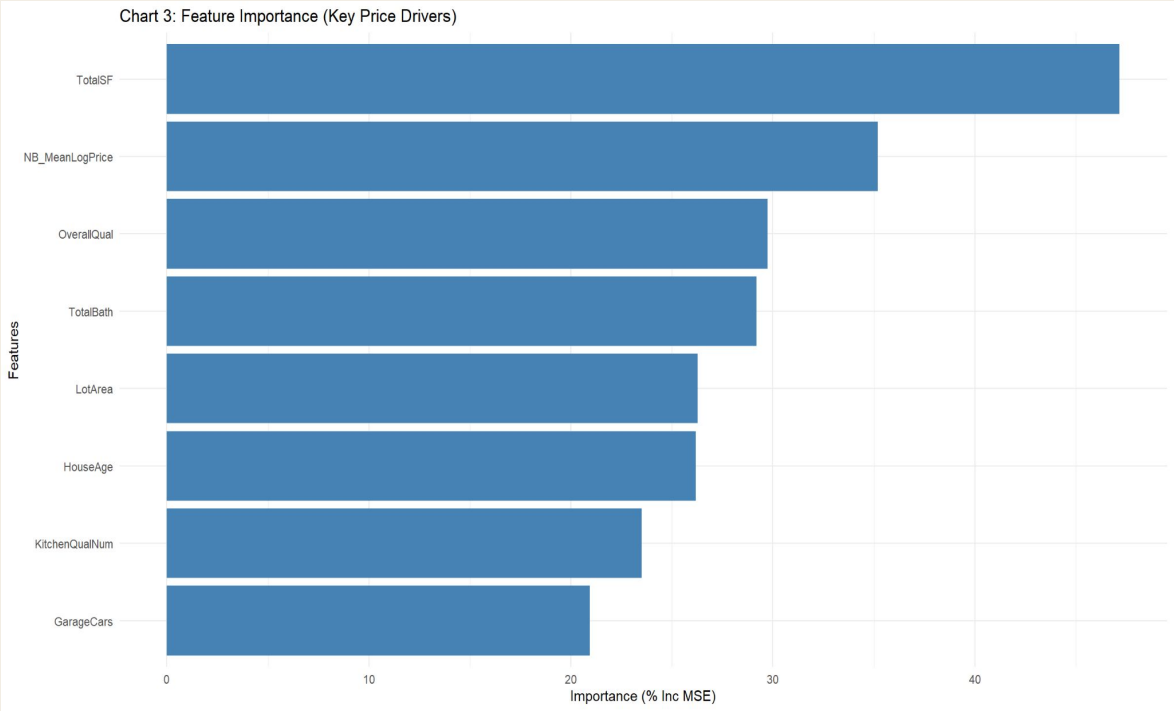
(Standardized)

Higher Error in Low-End & Luxury

Random Forest More Stable

Results

Feature Importance Analysis: Which features are most important for predicting house price



- **Size as Primary Driver**
- **Location Matters (Neighborhood Effect)**
- **Quality & Age Influence Price**

Conclusion and Future Work

- **Random Forest Superiority (Accuracy, Stability)**
- **Precision (6.5% MAPE)**
- **Luxury Pricing Paradox**
- **Human-in-the-Loop**

- **Adding Image Data (CNNs)**
- **Economic Factors (mortgage rates, local employment growth)**

ANY
QUESTIONS?

References

Boehmke, B., & Greenwell, B. (2020). *Hands-on machine learning with R*. https://bradleyboehmke.github.io/HOML/index.html

De Cock, D. (2011). Ames, Iowa: Alternative to the Boston housing data as an end of semester regression project. *Journal of Statistics Education, 19*(3). https://www.kaggle.com/datasets/shashanknecrothapa/ames-housing-dataset

Freddie Mac. (2026). *Primary mortgage market survey*. https://www.freddiemac.com/pmms

Ho, W. K., Tang, B., & Wong, S. W. (2021). Predicting property prices with machine learning algorithms. *Journal of Property Research*. https://www.mdpi.com/2220-9964/15/1/46

Kan, K., Kwong, S., & Leung, C. (2017). *The Zillow Prize competition*. https://www.kaggle.com/c/zillow-prize-1

National Association of Realtors. (2025). *First-time home buyer share falls to historic low of 21%*. https://www.nar.realtor/newsroom/first-time-home-buyer-share-falls-to-historic-low-of-21-median-age-rises-to-40

U.S. Census Bureau. (2025). *Income in the United States: 2024* (Report No. P60-286). https://www.census.gov/library/publications/2025/demo/p60-286.html

Zillow. (2026). *United States home values*. https://www.zillow.com/home-values/102001/united-states/